

The Metric-to-Fixed-Atom Selector Gate: A Singleton Obstruction, Exact Bridge Hypotheses, and the Pointwise Return

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Abstract

TPC-170 proves a fixed- X -power, all-prefix packet theorem for Lebesgue-almost every fixed phase on a prescribed determinant-two schedule. TPC-180 shows that the frozen source corpus does not yet contain the value-bearing production phase registry required to name the physical atom. We isolate the remaining selector question even conditionally on a future positive registry.

For the schedule-dependent bad sets E_n , Borel–Cantelli proves $\text{meas}(\limsup E_n) = 0$. It does not prove that a prespecified singleton α_* lies outside that limsup: the null set may be exactly $\{\alpha_*\}$. We formulate the minimal source-backed bridge contract, whose load-bearing theorem is the schedule-specific assertion $\alpha_* \notin \limsup E_n$. No artifact currently supplies this assertion. Therefore `H2.metric_fixed_atom_crosswalk` remains `NOT_TESTABLE`, and only the uncontrolled atomic promotion method receives a scoped stop. The pointwise bad-endpoint and direct additive-twist routes remain parent-ready and open. This is a rigorous L1 quantifier obstruction, not a program-positive L2 estimate.

Claim boundary. Lebesgue-almost-every fixed phase means that one phase is followed through the prescribed schedule; it does not identify a named atom. The scoped nonimplication below does not prove a large literal Möbius twist, stop a pointwise theorem, or stop an architecture. The metric powers $\delta < 1/4$ are not fixed-atom powers. No program-positive L2, strict $1/400$, prime-pair lower bound, or twin-prime theorem is proved.

1 The imported metric theorem

At scale n , let $\mathcal{P}_n$ be the packet list prescribed before its exceptional phases are read. For $p \in \mathcal{P}_n$, write

$$G_{n,p}(\alpha) = \frac{q_{n,p}}{T_{n,p}} \max_{k \leq L_{n,p}} |S_{n,p,k}(\alpha)|. \quad (1)$$

TPC-169 proves

$$\int_{\mathbb{T}} G_{n,p}(\alpha)^2 d\alpha \leq V_{n,p}, \quad (2)$$

where

$$V_{n,p} = D_{n,p}^2 \|\rho_{n,p}\|_{\infty}^2 \left(\frac{q_{n,p}}{T_{n,p}} + \frac{q_{n,p}^2}{T_{n,p}^2} \right) \quad (3)$$

[3]. For thresholds $\lambda_n > 0$, define

$$E_n = \left\{ \alpha \in \mathbb{T} : \max_{p \in \mathcal{P}_n} G_{n,p}(\alpha) > \lambda_n \right\}. \quad (4)$$

TPC-170 proves that

$$\sum_n \lambda_n^{-2} \sum_{p \in \mathcal{P}_n} V_{n,p} < \infty \quad (5)$$

implies

$$\text{meas}\left(\limsup_{n \rightarrow \infty} E_n\right) = 0. \quad (6)$$

Equivalently, for Lebesgue-almost every fixed α , every declared packet and every prefix is eventually at most λ_n [4].

Under the dyadic, polylogarithmic packet envelope, one may choose

$$\lambda_n = X_n^{-\delta}, \quad \delta < \frac{1}{4}. \quad (7)$$

The exact imported signature is

axis	proved value
carrier	EXPLICIT_PACKET_CORRIDOR
phase	LEBESGUE_AE_FIXED_PHASE
endpoint	ALL_PREFIX_THETA_SHELL
scale	EVENTUALLY_PRESCRIBED_SCHEDULE
decay	FIXED_X_POWER_PHASE_METRIC
support	ACTUAL_CORE.

(8)

The word “fixed” in the phase row means that the same α is followed across n . It does not change the almost-everywhere quantifier to evaluation at a named atom.

2 The two independent missing inputs

TPC-180 audits the value-bearing phase sources and obtains

$$\begin{aligned} \text{H9.phase_cell_registry} &= \text{NOT-TESTABLE}, \\ \text{named_physical_atom_id} &= \text{null}, \quad \text{phase_value_mod_1} = \text{null}. \end{aligned} \quad (9)$$

The fixed- $h_0 = 2$ data fact remains proved, but is not a phase value [7].

Even if a later source supplies $\alpha_\star$, one more theorem is needed:

$$\boxed{\alpha_\star \notin \limsup_{n \rightarrow \infty} E_n} \quad (10)$$

for the exact production schedule and bad sets (4). The registry is a data input; (10) is an arithmetic or selector theorem. Their roles and evidence cannot be merged.

Definition 2.1 (Legal metric-to-fixed-atom bridge). A legal bridge contains:

- (i) the exact source-located named atom and phase value;
- (ii) a hash and source locator for the exact production packet schedule;
- (iii) the schedule’s precise sets E_n ;
- (iv) a theorem proving (10);
- (v) the same fixed $\alpha_\star$ at every scale; and
- (vi) coverage of every production packet to which the conclusion is applied.

A source hash certifies only identity. It does not prove item (iv). Nor do a phase- L^2 estimate, density-one control on a separated registry, a phase selected after the bad set, or phases α_n that drift with n [1, 2, 5, 6].

3 The singleton obstruction

Proposition 3.1 (Full measure does not select a singleton). *Let $\alpha_\star \in \mathbb{T}$ be prescribed. From $\text{meas}(G) = 1$ alone one cannot infer $\alpha_\star \in G$.*

Proof. Take

$$N = \{\alpha_\star\}, \quad G = \mathbb{T} \setminus N.$$

Then $\text{meas}(N) = 0$ and $\text{meas}(G) = 1$, but $\alpha_\star \notin G$. $\square$

Corollary 3.2 (Borel–Cantelli selector nonimplication). *Equation (6), without an independent theorem about $\alpha_\star$, does not imply (10).*

Proof. Apply [proposition 3.1](#) with

$$N = \limsup E_n, \quad G = \mathbb{T} \setminus \limsup E_n.$$

The metric theorem controls the measure of N , not membership of a prespecified singleton. $\square$

TPC-170 also realizes the logical sharpness with constant coefficients at phase zero. That example is not asserted to be the literal Möbius packet. The general obstruction in [proposition 3.1](#) is a quantifier nonimplication, not a coefficient lower bound.

Theorem 3.3 (Exact scoped stop). *The implication*

$$\text{LEBESGUE_AE_FIXED_PHASE} \implies \text{NAMED_FIXED_ATOM} \tag{11}$$

is invalid without a bridge satisfying [definition 2.1](#). Therefore the arithmetic method

$$\text{phase_metric_uncontrolled_atomic} \tag{12}$$

has status STOP-SCOPED in scope UNCONTROLLED_ATOMIC_PROMOTION_ONLY.

Proof. [Corollary 3.2](#) disproves the inference at the only step where the almost-everywhere theorem is promoted to the named atom. Adding a source-backed proof of (10) changes the premise and lies outside the uncontrolled method. A direct pointwise theorem also changes the phase quantifier rather than using (11). Hence the stop has exactly the stated scope. $\square$

4 Selector verdict and sufficient bridge hypotheses

The present state fails both inputs to [definition 2.1](#): the value-bearing registry in (9) is absent, and no schedule-specific avoidance theorem (10) is source-locked. Thus

$$\begin{aligned} \text{H2.metric_fixed_atom_crosswalk} &= \text{NOT-TESTABLE}, \\ \text{SelectorConstructed} &= \text{false}, \\ \text{FixedAtomDecayObtained} &= \text{false}. \end{aligned}$$

(13)

Three forms of genuinely new input could change a future snapshot:

- (a) a theorem proving (10) for the exact named phase and exact schedule;
- (b) a source-compatible point-evaluation or continuity estimate strong enough to dominate $G_{n,p}(\alpha_\star)$ by controlled values with a summable error; or
- (c) an independent pointwise fixed-phase theorem, which exits the metric bridge and enters a pointwise route.

Merely declaring the named phase “generic” is not one of these hypotheses unless a theorem with the exact schedule-dependent quantifiers is supplied.

5 Return to the two pointwise routes

TPC-171 and TPC-172 keep two targets parent-ready and open:

$$\mathcal{F}_{\text{point}} = \{\text{O161.bad_endpoint_pointwise_fixed_atom}, \text{O161.direct_additive_twist_fixed_atom}\}. \quad (14)$$

Both require the physical signature

phase	NAMED_FIXED_ATOM	(15)
endpoint	DETERMINISTIC_ALL_PREFIX	
scale	DETERMINISTIC_ALL_SCALE	
decay	FIXED_X_POWER_FIXED_ATOM	
support	ACTUAL_ACTIVE_SUPPORT.	

Proposition 5.1 (Pointwise frontier preservation). *The scoped stop in [theorem 3.3](#) does not change either element of $\mathcal{F}_{\text{point}}$ from OPEN_PARENT_READY, and it does not stop or reroute an architecture.*

Proof. The stopped node has arithmetic-negative role and the exact uncontrolled-atomic-promotion scope. The nodes in (14) are independent arithmetic targets whose hypotheses assert pointwise fixed-atom estimates. Neither depends on the invalid implication (11). Architecture nodes have a disjoint route kind. Therefore the scoped stop has no typed edge that could close either pointwise target or an architecture. $\square$

The correct route decision is consequently

route cell	state	(16)
source-backed metric bridge	NOT-TESTABLE	
uncontrolled atomic promotion	STOP-SCOPED	
bad-endpoint pointwise fixed atom	OPEN_PARENT_READY	
direct additive twist fixed atom	OPEN_PARENT_READY	
architecture reroute	false.	

6 Level ledger and reproducible audit

TPC-181 separates:

- L0: source hashes, strict schemas, output regeneration, and finite mutation diagnostics;
- L1: [propositions 3.1](#) and [5.1](#) and [theorem 3.3](#), which are rigorous quantifier and typed-route statements; and
- L2: none.

The metric exponent $\delta < 1/4$ in (7) remains in the phase-metric ledger. It is explicitly ineligible as fixed-atom power.

The generator writes

```
experiments/tpc181_selector_gate.json
experiments/tpc181_selector_gate_audit.json
```

and records the selector verdict, scoped stop, named-atom status, metric quantifier, both O161 routes, and claim boundary. Mutation regressions reject almost-everywhere promotion, fabrication of the missing registry, charging metric power as fixed-atom power, using the scoped stop against a pointwise target or architecture, and a scale-dependent selector.

7 Conclusion

The metric route ends at a precise theorem interface. TPC-170 gives full-measure eventual control on its exact prescribed schedule; TPC-181 proves that this measure statement cannot select a prespecified singleton. Since TPC-180 also lacks the named value and locator, the source-backed selector is not testable. The failure is method-scoped, so the program returns to the two pointwise O161 targets without relabeling them as stopped and without claiming any new L2 progress.

References

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